NOTICE OF
REACTIVATION

INCH-POUND

MIL-I-60277A
NOTICE 3
26 November 2003
SUPERSEDING
NOTICE 2
10 July 1999

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M26 ASSEMBLY

MIL-I-60277A, dated 3 January 1989, is hereby reactivated and may be used for either new or existing design acquisition.

Custodians
Navy - OS
Air Force - 99

Preparing activity:
Navy - OS
(Project 1377-0210)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

AMSC N/A

FSC 1377

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INCH-POUND

MIL-DTL-60277B
26 November 2003
SUPERSEDING
MIL-I-60277A
3 January 1989

DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M26 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the Cartridge Actuated Initiators M26 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-A-2550	Ammunition, General Specification For
MIL-I-60508	Initiator, Cartridge Actuated, Delay, M6A1, M26, M33, M44, Metal Parts Assembly

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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MIL-I-82760	Initiator, Cartridge Actuated, Delay, M6A1, M26, M33, Remanufactured Metal Parts Assembly

STANDARDS

MILITARY

MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DOD Preferred Methods for Acceptance of Product
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

8596123	Cartridge, Delay, M90 Assembly
8796826	Box, Fiberboard (Inner) Packing, Ammunition for Initiators, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1 or M49A1
8796827	Box, Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1, or M49Al in Fiber Container, M375
DL8597941	Initiator, Cartridge Actuated, Dela y, M26, Assembly
9216824	Box, Fiberboard (Outer) Alternative, Packing Ammunition, for Initiators, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1, or M49A1

U.S. AIR FORCE

TPO F00-138-3770	Initiators

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC A8402	Quality Management and Quality Assurance

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(Copies of this document are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Inspection

(Copies of this document are available online at http://www.astm.org/cgi-bin/SoftCart.exe/STORE/standardsearch.shtml?L+mystore+afpt6301+1062008944 or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Primary components. For the purpose of this specification, the M90 delay cartridge (Drawing 8596123) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.3 Breakaway torque. The chamber and cap of each assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.2.3.1).

3.4 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction using the procedures in 4.3.2.

3.5 Functional.

3.5.1 Ballistics. Initiators loaded with cartridges from one accepted lot and temperature conditioned shall perform within the pressure values and delay time specified in table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, assembly number MS28741-4, and a pressure chamber with a volume of 1 cubic inch (see figure 1).

3.5.1.1 Initiation. The initiator shall be fired by a source delivering an output pressure of 700 +25, -0 pounds per square inch (gage) (psig) applied at a rate between 10,000psig/sec and 50,000psig/sec at 70 ± 5°F.

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TABLE I. Pressure values.

Temperature (°F)	Pressure (psi)		Delay time (sec) (see 6.8.2)
	Minimum	Maximum	
-60 to –70	1300	2200	0.350 ± 0.100
65 to 75	1300	2200	0.300 ± 0.050
195 to 205	1300	2200	0.250 ± 0.100

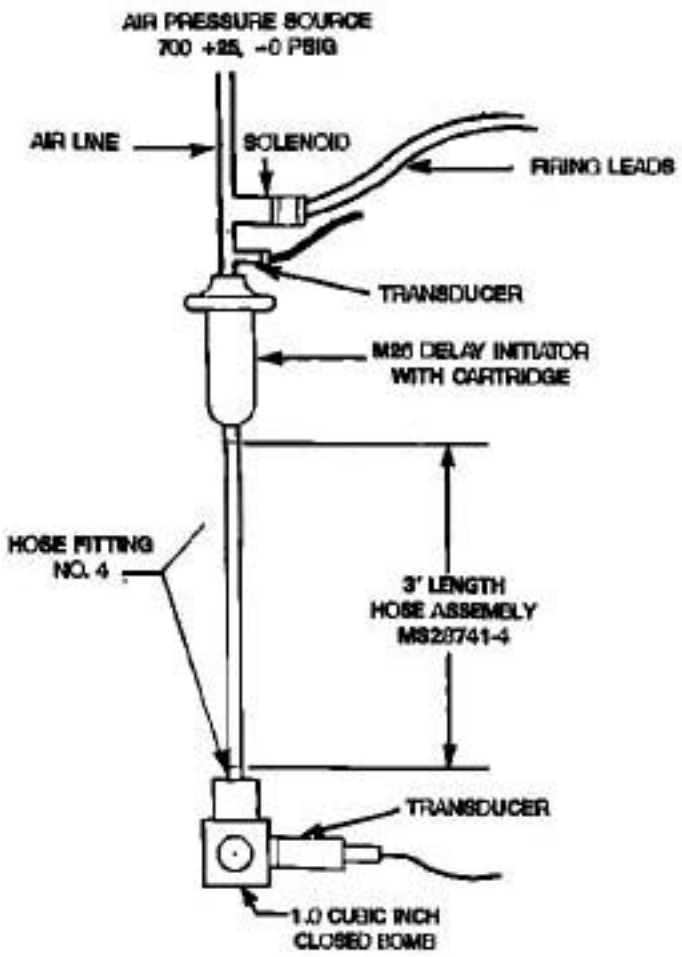


FIGURE 1. Performance test layout (top view).

3.5.2 Misfires. There shall be no misfires.

3.5.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.8.1).

3.6 Workmanship. Initiators shall be free of cracks, splits, incomplete or damaged threads on inlet and outlet ends, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.6.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

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3.6.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

3.7 Radiographic examination. Each initiator assembly shall be radiographically examined for defects as specified in 4.4.

3.8 Component parts. All component parts shall meet the applicable requirements of MIL-I-60508 or MIL-I-82760 prior to assembly.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection

4.1.1 Quality assurance terms and definitions. Reference shall be made to ASQC A8402 to define the quality assurance terms used.

4.1.2 Inspection. Inspection shall be in accordance with MIL-A-2550 and ASTM E1742.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting activity formation and identification of lots shall be in accordance with MIL-STD-1916.

4.2.1.2 Size of lot. A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with one lot of accepted cartridges.

4.2.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects, using the sampling plan and associated criteria specified in the contract (see 6.2 and 6.3). All nonconforming material shall be rejected. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability.

4.2.2.1 Cartridge, Delay, M90, Drawing 8596123, covering a detail of Drawing 8597941.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	O-ring missing	Visual
4	Sealing disc missing, damaged, or improperly secured	Visual
5	Marking missing, incorrect, or illegible	Visual

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Characteristic	Defect	Method of Inspection
Major		
101	O-ring split, cut, or cracked	Visual
102	Foreign matter present	Visual
Minor		
201	Case scratched, wrinkled, bulged, or dented	Visual
202	Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing 8597941, less Drawings 8594745, 8595254, 8593313, 8593314, and 8596123.

Critical		
1	Protrusion of firing pin	Gage
2	O-ring missing	Visual
Major		
None defined		
Minor		
None defined		

4.2.2.3 Assembly, Drawing 8597941, less Drawings 8595254, 8593314, 8593313, and 8596123, and specifications MIL-C-5501/10 and MIL-C-5501/11.

Critical		
1	Shear pin not engaged in firing pin	Visual/Manual
2	Shear pin set screw missing	Visual
Major		
101	O-ring missing or not lubricated	Visual
102	Thread damaged	Visual
103	Foreign matter present	Visual
Minor		
None defined		

4.2.2.4 Assembly, Drawing 8597941, less Drawings 8595264 and 8593314, and specifications MIL-C-5501/10 and MIL-C-5501/11.

Critical		
1	Cartridge loose	Visual/Manual
2	Cartridge retainer loose	Visual/Manual
3	Metering slot of cartridge retainer missing	Visual
4	Elongated slot of cartridge retainer missing	Visual
Major		
None defined		

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Characteristic	Defect	Method of Inspection
	Minor	
	None defined	

4.2.2.5 Assembly, Drawing 8597941, less specifications MIL-C-5501/10 and MIL-C-5501/11.

	Critical	
1	Cartridge retainer position improper	Gage
2	Foreign matter in gas port	Visual
	Major	
101	Length from cap boss face to gas port face, max	Gage
102	Set screw missing, loose or above flush	Visual/Manual
	Minor	
	None defined	

4.2.2.6 Assembly, Drawing 8597941.

	Critical	
1	Base of firing pin housing not in complete contact With face of cartridge	Manual/X-ray
2	Any part missing, except as otherwise classified	Visual/X-ray
3	Improper assembly	Visual/X-ray
4	Marking missing, Incorrect, or Illegible	Visual
	Major	
101	Height of cap, max	Gage
102	Set screw not covered with barrier material	Visual
	Minor	
201	Total length, max	Measure
202	Location of labels Incorrect	Visual
203	Protective cap or plug missing	Visual
204	Protective finish inadequate or defective	Visual

4.2.2.7 Box, Fiberboard (Inner) Ammunition, Drawing 8796826.

	Critical	
	None defined	
	Major	
	None defined	
	Minor	
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

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4.2.2.8 Box, Ammunition Packing, Drawing 8796827.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.2.2.9 Box, Fiberboard (Outer) Alternative Ammunition Packing, Drawing 9216824.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with the requirements of 3.3 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with the requirement of 3.4 shall be cause for rejection of the initiator.

4.2.3.3 Functional tests. The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics. Twenty-one units shall be tested to determine compliance with the requirements of 3.5. The units shall be temperature conditioned as provided in table II and 4.3.3.2. Failure of any unit to comply with the requirements of 3.5 shall be cause for rejection of the lot.

4.2.3.3.2 Retest. There shall be no retests.

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4.2.3.3.3 Test Failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

TABLE II. Conditioning temperature.

Temperature (°F)	Sample size
-60 to -70	7
65 to 75	7
195 to 205	7

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench set at the torque requirement specified in 3.3 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in a suitable vise, and the torque shall be applied to the cap.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.3.3 Functional.

4.3.3.1 Ballistics. The test sample shall be rigidly supported in the fixture and fired by a source delivering an output of 700 + 25, - 0 psig applied at a rate between 10,000psig/sec and 50,000 psig/sec at 70 ± 5° F (see figure 1). The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperature.

4.3.3.2 Temperature conditioning. Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning initiators at –65 ± 5 °F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber. Required pressure readings shall be taken as specified in 3.5.1.

4.4 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with ASTM E1742, with exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

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a. Pin, firing P/N 8594744
b. O-ring MS28775-214
c. Housing, firing pin P/N 8594858
d. Retainer, cartridge P/N 8593313
e. O-ring MS28775-009
f. Pin, shear P/N 8594745
g. Shear pin screw —
h. Set screw, initiator cap —
i. Cartridge, Delay M90 P/N 8596123

4.5 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD orin-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The initiator was designed specifically for military use and has no commercial application.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.4).
c. Production lot size and test activity (see 4.2.1).
d. Inspection sampling plan and acceptance/rejection criteria (see 4.2.2 and 6.3).
e. Direction for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.4).
f. Level of marking and packing (see 5.1, and 6.7).
g. Safety precautions (see 6.5).

6.3 Inspection sampling plan and acceptance/rejection criteria. The sampling plan and acceptance/rejection criteria of table III has been used successfully in procuring, testing, and examining the M26 Cartridge Actuated Initiator. The total number of allowable defects per class for either the major or minor classification should not exceed the acceptance number of table III. The contracting activity is cautioned that any deviation from the table III values should be warranted and verified statistically considering the criticality of the initiator's function.

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TABLE III. Sampling plan.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 — 300	100% examination	—	—	—	—
301 — 800	50	0	1	1	2
801 — 1800	90	1	2	2	3

6.4 Explosive hazard classification information.

Hazard Classification/Division: 1.4
Storage Compatibility Group: C
DOT Class: C
DOT Marking - Explosive Power Device - Class C - Handle Carefully - Keep Fire Away

6.5 Safety precautions. The safety precaution requirements of "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.6 Hazard notice. The initiator described herein and certain of its components are flammable, explosive, or both. Consequently, it presents hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically-induced incidents.

6.7 Packaging

6.7.1 Packing and marking. Packing and marking should be levels A or B as specified by the contracting activity:

Level A - In accordance with Drawings 8796826 and 8796827
Level B - In accordance with Drawings 8796826 and 9216824

6.7.1.1 Lot numbers. Lot numbers should be in accordance with MIL-STD-1168.

6.7.2 Air Force packaging. Air Force packaging should be in accordance with TPO F00-138-3770.

6.8 Definitions.

6.8.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.8.2 Delay time. Delay time is defined as the time from primer impact to the start of pressure rise.

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6.9 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes. The changes from the previous issue were authorized by ECP number 0352201010.

Custodian
Navy - OS
Air Force – 99

Preparing activity
Navy - OS
(Project 1377-0211)

Review activity
Air Force - 70

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